

Pokhara University
Faculty of Management Studies

Course Code: MTH113

Full marks: 100

Course title: **Mathematics I**

Pass marks: 45

Nature of the course: Theory

Credit: 3 Hrs.

Year: First, Semester I

Total periods: 48 hours

Level: Bachelor

Program: BCSIT

1. Course Description

This course covers basic logic, set theory, the real number system, functions and graphs, limit and continuity, derivatives and their applications, integration, matrices and determinants, and permutations and combinations.

2. Course Objective:

The course is designed with the following objectives:

- To acquaint the students with fundamental mathematical concepts with a focus on how they apply to business, economics, and information technology.
- To develop skills among the students that they need to effectively apply mathematical techniques to real-world problems.

3. Method of Instructions

General Instructional Technique: Lecture, Discussion, Readings and Question Answer

Specific Instructional Technique: Self-Directed Learning and Case Study

4. Course Detail with Specific Objectives

Specific Objectives	Content
• Comprehend the idea of logic and able to use it to support mathematical claims as well as practical issues. • Use various set operations to solve the specific problem and determine the number of items in a given set.	Unit 1: Basic Concept [9 Hrs.] 1.1. Elementary Logic 1.2. Connectives, Quantifiers, 1.3. Basic laws of logic, 1.4. Quantifiers 1.5. Techniques of proof. 1.6. Sets, Types of sets, Venn diagram,

	1.7. Set operations, Laws of the algebra of sets (without proof), 1.8 Real number system, Representation of real numbers on the Real line, 1.9 Properties of real numbers (without proof), ordered sets, Inequalities, 1.10 Intervals, Absolute value, Cartesian product, Relation.
Specific Objectives	Contents
- Identify the dependent and independent variables to find the domain and range of the function. - Explain the concept of limit and continuity and use it in real-life problems.	Unit 2: Functions, Limit, and Continuity [8 hrs.] 2.1. Constants and variables, Concept of functions, 2.2 Domain, and range of a function, Types of functions (algebraic, logarithmic, Trigonometric, and exponential functions), 2.3. Graphic representation 2.4 Application of functions to business and economics. 2.5 Limit of a function, properties of Limit, Indeterminate Forms, 2.6 Limits of Polynomial and Rational Functions, Limits at infinity, 2.7 Continuity, Continuity at a Point, 2.8 Business Application of Limit.
Specific Objectives	Contents
- Identify the rate change of a function at a particular point. - Find the derivative of algebraic, exponential, logarithmic, and trigonometric functions.	Unit 3: Derivative [6 Hrs.] 3.1 Derivative 3.2. Average rate of change 3.3. Derivative as a slope of the tangent to curves 3.4. Methods of differentiation (power rule sum rule, product rule, quotient rule and chain rule) 3.5 Differentiation of implicit and parametric functions 3.6 Derivative as a rate of change. Higher order derivatives.

Specific Objectives	Contents
- Establish the relationship between the derivative and concavity of the function. - Identify the increasing and decreasing function. - Find the maxima minima of a function, using the first and second-order derivative.	Unit 4: Application of Derivatives [6 Hrs.] 4.1. Increasing and decreasing functions 4.2. Critical point, Point of inflection 4.3. Maximum and minimum value of the function 4.4. Marginal analysis in Business and Economics 4.5. Concavity of the function, Marginal Profit Analysis 4.6. The mean value theorem, Optimization problem
Specific Objectives	Contents
- Use different Rules and techniques of integration to solve the related problems. - Identify the value of the function from the corresponding marginal value.	Unit 5: Integrals [6 Hrs.] 5.1 Indefinite Integral (algebraic, exponential, logarithmic, and trigonometric functions) 5.2 Techniques of Integration (Integration by simplification, Substitution method, Integration by parts) 5.3 Definite integral, 5.4 Properties of the definite integral
Specific Objectives	Contents
- Perform matrix operations and solve related problems - Evaluate the value of the determinant and use the concept of determinant for solving linear equations.	Unit 6: Matrices and Determinants [7 Hrs.] 6.1. Introduction 6.2. Types of matrices 6.3. Matrix operations 6.4. Transpose of a matrix, Determinant of a matrix, Minors, and cofactors of the matrix 6.5. Properties of determinants (Singular and non-singular matrix)

	6.6.Solution of a system of linear equations having a unique solution of up to three variables (Cramer's rule).
Specific Objectives	Contents
- Identify the difference between permutation and combination. - Use the concept of permutations and combinations in real-life problems.	Unit VII: Permutations and Combinations [6 hrs] 7.1 Basic principles of counting, factorial notation, 7.2. permutation, Permutation of objects alike, 7.3. Permutation with restrictions, Circular permutation, 7.4 Combination, and Combination with restrictions.

5. Evaluation System and Student's Responsibilities

Evaluation System

In addition to the formal exam(s), the internal evaluation of a student may consist of quizzes, assignments, lab reports, projects, class participation, etc. The tabular presentation of the internal evaluation is as follows.

External Evaluation	Marks	Internal Evaluation	Weight	Marks
Semester-End examination	50	Theory		50
		Attendance & Class Participation	10%	
		Assignments	20%	
		Presentations/Quizzes	10%	
		Internal Assessment	60%	
Total External	50	Total Internal		50
Full Marks: 50 + 50 = 100				

6. Student's Requirement

Each student must secure at least 45% marks separately in both internal assessment and practical evaluation with 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such a score will be given NOT QUALIFIED (NQ) to appear for the Semester-End Examinations. Students are advised to attend all the classes, formal exams, tests, etc., and complete all the assignments within the specified time period. ***Students are required to complete all the requirements defined for the completion of the course.***

7. Prescribed Books and References

Prescribed Books

1. Budnick, F. (2017). *Applied Mathematics for Business, Economics and the Social Sciences* (4 ed.). McGraw-Hill Higher Education.
2. Pahari, N. P., & Shrestha, R. M. (2013). *Fundamentals Of Mathematical Analysis* (1 ed.). Sukunda Pustak Bhawan.

References

1. Thomas, G. B., & Finney, R. L. (1995). *Calculus and Analytic Geometry* (9 ed.). Addison Wesley.
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